

From Gross Edge to Net Profit

Fill Timing and Trading Costs in an Intraday Reversal Strategy

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Abstract

A 5-minute direction model on 431 S&P 500 stocks has a steady gross edge: filled at the close of the bar that produced the signal, \$1 million grows to \$4,091,721 in 4.05 years with a Sharpe ratio of 9.01 and a worst drawdown of -2.2% . The present paper asks how much of that survives execution. We make three contributions. First, we prove that the difference between filling at the signal bar's close and at the next bar's open equals position change times the jump between the two prints, and show that this jump is bid-ask bounce: it removes 46% of the gross edge. Second, we measure trading costs per stock and per day from the bars themselves, including an exact lower bound: no market order can pay less than half a \$0.01 tick. Third, we test whether trading less, trading tight-spread stocks, or combining timeframes rescues the strategy. None of the three does. Every one of the 46 versions tested loses money at the lower-bound cost; the best out-of-sample version turns \$1 million into \$870,953 over 2022–2025. The edge grows with the spread, but a follow-up test shows that limit orders resting at the last price do not collect it: their fills are adversely selected.

1 Introduction

Motivation. Most strategies that look good in a backtest lose money when traded. For high-frequency strategies the main reason is simple: they trade often, and each trade costs money. Novy-Marx and Velikov (2016) show that trading costs remove most of the profit of many published anomalies with high turnover. Intraday strategies turn over their whole book many times a day, so a cost of a fraction of a basis point per trade decides the result. Yet backtests often ignore costs or charge one flat number, and often assume a fill at the very price that produced the signal.

Reversal and the spread. Short-horizon returns reverse (Lehmann, 1990). Part of this is bid-ask bounce: a trade at the bid followed by a trade at the ask looks like a reversal even when the value has not changed (Roll, 1984). Bounce explains much of short-horizon reversal (Kaul and Nimalendran, 1990; Jegadeesh and Titman, 1995), and reversal profits can be read as the return to providing liquidity (Nagel, 2012). A reversal strategy measured on traded prices may therefore be measuring the spread, which a liquidity taker pays rather than earns.

This paper. We take a model with a real and steady gross edge, a multi-timeframe ARX on 5-minute bars described in Section 2, and follow its profit from the backtest to what a trader would actually get. We ask three questions. How much of the gross edge depends on the assumed fill price? What does the strategy cost to trade, measured per stock and per day? And can any version, by trading less or trading only cheap stocks, pay for itself?

Findings.

1. **About half the gross edge is the fill assumption** (Section 3). Moving the fill from the signal bar’s close to the next bar’s open, one print later, removes 46% of the 5-minute edge. We prove the difference equals position change times the jump from the close to the next open, and show that this jump reverses the last bar’s move in 94% of stocks after down bars.
2. **Every version loses money at the smallest possible cost** (Section 5). Half a \$0.01 tick is a cost no market order can avoid. At that cost all 46 versions lose money.
3. **The edge is the spread.** Gross breakeven rises from 0.066 to 0.197 bps from the tightest to the widest spread quintile, and correlates 0.63 with each stock’s spread.
4. **The result is statistically solid.** Bootstrap confidence intervals for the net return lie below zero; the gross edge survives a correction for 102 trials (deflated Sharpe 0.978).

2 The strategy in brief

The paper stands alone, so the strategy of the companion paper is summarised first.

Forecast. For each stock and each of six timeframes (5 minutes to 4 hours), a linear regression forecasts the next bar’s close-to-close return from five lags of return and five lags of a bounded bar-level flow proxy, $\tanh(\text{sign}(C - O) \cdot |C - O| / (H - L) \cdot \log(1 + V))$. Each forecast is divided by recent volatility, bounded with $\tanh$, and the six scores are averaged, with equal weights or with weights learned on 2021 only. Models are refitted monthly on the previous year.

Positions. Long one unit if the score is positive, short if negative (“sign”); or a position equal to the score (“size”); or the sign only when the score is in the top quarter of its recent distribution, holding the previous position otherwise (“filtered”). “CLOSE 5m” is the 5-minute timeframe alone.

Book and period. 431 stocks with 5-minute regular-hours bars, equal weight, 426.6 stocks per bar on average, 2020-12-18 to 2025-01-10 (4.05 years); 2022–2025 is a hold-out for the learned weights. Positions are closed at every session end. A look-ahead test, run on truncated data and shown to catch a deliberate leak, confirms that no decision uses later data.

What the signal is. The coefficient on the latest return is negative in 76% of monthly refits; the flow coefficients are coin flips. The model is a one-bar reversal.

3 Fill timing

A backtest must state the price at which each trade is filled. That choice alone changes the result by about half, so it is settled before costs are measured.

3.1 Three fill rules

A decision is made at the end of 5-minute bar t , with position w_t held into bar $t + 1$. We compare:

- **Close fill:** trade at C_t , the last price of the signal bar. Profit $w_t (C_t \rightarrow C_{t+1})$. The close fill is the usual backtest convention.

- **Next-open fill:** trade at O_{t+1} , the first print after the signal, which is what a market order sent at the bar close receives. The old position w_{t-1} is held from C_t to O_{t+1} , the new one from O_{t+1} to C_{t+1} .
- **One-bar delay:** trade at C_{t+1} , a signal one bar stale.

Returns are in bps of log price, so $(C_t \rightarrow C_{t+1}) = (C_t \rightarrow O_{t+1}) + (O_{t+1} \rightarrow C_{t+1})$ exactly.

Proposition 1 (What the close fill adds). *Per bar, profit with the close fill minus profit with the next-open fill equals $(w_t - w_{t-1}) \cdot (C_t \rightarrow O_{t+1})$.*

Proof. Close fill: $w_t(C_t \rightarrow O_{t+1}) + w_t(O_{t+1} \rightarrow C_{t+1})$. Next-open fill: $w_{t-1}(C_t \rightarrow O_{t+1}) + w_t(O_{t+1} \rightarrow C_{t+1})$. Subtract. $\square$

The close fill therefore earns extra exactly when the position changes in the direction of the jump from the close to the next print. For a reversal strategy the position changes after a bar moved: after a down bar it buys. If the close of a down bar tends to be printed at the bid, the next print is higher, and the close fill credits the strategy with a gain that the buyer, who pays the ask, never gets.

The jump is bounce. The move from the close to the next print averages -0.112 bps after up bars and $+0.148$ bps after down bars: it goes against the last bar in 87% and 94% of stocks respectively. The first print after the close moves back towards the middle of the spread, as Roll (1984) predicts for prints that alternate between bid and ask.

Table 1: Breakeven half-spread (bps) under each fill rule, and gross Sharpe ratio. Breakeven is gross profit per bar divided by turnover per bar.

Strategy	Breakeven (bps)			Gross Sharpe	
	Close fill	Next open	1-bar delay	Close fill	Next open
CLOSE 5m	0.190	0.103	0.030	9.01	4.99
CLOSE MTF, equal, sign	0.318	0.182	0.086	6.24	3.61
CLOSE MTF, equal, filtered	0.590	0.280	0.125	2.76	1.32
CLOSE 4h only	0.566	0.558	0.577	1.46	1.44
MID 5m	-0.243	-0.133	-0.057	-7.35	-4.14
<i>Out of sample, Jan 2022 – Jan 2025</i>					
CLOSE MTF, learned, filtered	0.563	0.219	0.095	2.07	0.81

Reading Table 1. For CLOSE 5m, the close fill earns 0.181 bps per bar and the next-open fill 0.098; the difference, 0.083 bps, is the term of Proposition 1, and is 46% of the close-fill edge (53% for the filtered book). Across strategies the breakeven falls by 43%–65%. With a one-bar delay almost nothing is left: the signal’s information lasts about one bar. The 4-hour model is the exception: it trades rarely, so the fill hardly matters. All results below use the next-open fill.

4 Trading costs

Whether the gross edge survives depends on the cost of each unit traded, so costs are measured per stock and per day from data available before each day.

4.1 Net profit and breakeven

With half-spread c (in bps) and turnover $|w_t - w_{t-1}|$, the net profit of a bar is gross profit $-c \cdot |w_t - w_{t-1}|$. A switch from long to short counts two and pays two half-spreads. The breakeven half-spread is gross profit per bar divided by turnover per bar: the largest cost at which the book still breaks even. A high Sharpe ratio built from small bets can coexist with a very low breakeven, so breakeven, not Sharpe, decides whether a strategy can be traded.

4.2 Three cost measures

We have no quote data, so costs are measured from the bars. All three measures are computed per stock and per day from data before that day.

Tick floor (exact lower bound). Prices move in \$0.01 ticks, so the quoted spread is at least one tick and a market order pays at least half a tick: $c^{\text{tick}} = 0.005/P \times 10^4$ bps, with P the previous session’s close. The tick floor is not an estimate. If a strategy loses money at this cost, it loses money at every real cost.

Abdi–Rinaldo (central estimate). Abdi and Rinaldo (2017) note that the distance between the close and the midpoint of the range, $c_t - \eta_t$ with $\eta = (h + l)/2$ in logs, contains half the spread with a random sign, and that this sign is independent across bars. Hence $S^2 = 4 \mathbb{E}[(c_t - \eta_t)(c_t - \eta_{t+1})]$. We average the products over the previous 20 sessions and take the square root (zero if negative). The estimator uses the close, so it measures exactly the kind of bounce a reversal strategy trades on.

Corwin–Schultz (upper estimate). Corwin and Schultz (2012) use the fact that the high–low range of one bar contains the variance once and the spread once, while the range over two bars contains the variance twice and the spread still once. Two equations give the spread. We apply it to consecutive 5-minute bars, set negative values to zero and average over the previous 20 sessions. The estimator assumes the price moves continuously within the bar, which fails for short, thinly traded bars and biases it upwards when volatility is high.

Table 2: Half-spreads across stocks (bps), average over the sample for each stock.

Measure	10%	25%	Median	75%	90%
Tick floor (lower bound)	0.15	0.25	0.47	0.81	1.36
Abdi–Rinaldo (central)	0.98	1.25	1.59	2.05	2.85
Corwin–Schultz (upper)	1.59	1.90	2.31	2.83	3.71

Reading Table 2. Liquid large caps usually quote one or a few ticks wide. The Abdi–Rinaldo median of 1.59 bps is a few times the tick floor, which is plausible. The Corwin–Schultz median of 2.31 bps is too high for stocks this liquid; its largest values are for the most volatile names (ENPH, CCL, MRNA), consistent with its volatility bias. We therefore base the verdict on the tick floor, use Abdi–Rinaldo as the central case, and report Corwin–Schultz only as an upper estimate. The rule was fixed before the cost results were seen.

5 Results

The results test the possible rescues in turn: trading tight-spread stocks, trading less, and combining timeframes, followed by the statistical confidence in the loss.

5.1 The edge grows with the spread

Table 3: Stocks split each day into quintiles of their past Abdi–Ranaldo spread (q1 tightest). Next-open fill. The split was fixed before the results were seen.

	q1	q2	q3	q4	q5
<i>Gross breakeven, next-open fill (bps)</i>					
CLOSE 5m	0.066	0.049	0.086	0.114	0.197
MTF, equal, filtered	0.135	0.095	0.126	0.452	0.553
MTF, learned, filtered	0.113	0.226	0.241	0.365	0.620
<i>Total return after the tick floor</i>					
CLOSE 5m	−73%	−44%	−54%	−53%	−54%
MTF, equal, filtered	−8%	−4%	−5%	−2%	−2%
<i>Total return after Abdi–Ranaldo costs (the ranking variable)</i>					
CLOSE 5m	+1%	−67%	−90%	−97%	−100%
MTF, equal, filtered	+1%	−7%	−15%	−19%	−31%

Reading Table 3. Gross breakeven is lowest in the tightest-spread stocks and highest in the widest: 0.066 to 0.197 bps for CLOSE 5m, 0.135 to 0.553 for the filtered book. Across stocks, breakeven correlates 0.63 with the stock’s spread. A signal that is strongest where the spread is widest is harvesting the spread.

Trading only cheap stocks does not help. Under Abdi–Ranaldo costs the tightest quintile roughly breaks even (+1% for CLOSE 5m). That near-breakeven is not credible: the same noisy estimate chooses the stocks and sets their cost, so stocks whose spread is under-estimated are chosen and then under-charged. At the tick floor, which has no estimation error, the same quintile returns −73%.

Per stock. Only 11% of stocks have a 5-minute breakeven above their own tick floor (37% for the filtered book), and 0% (6%) above their own Abdi–Ranaldo spread.

5.2 Performance in dollars

Dollar paths show the size of the fill and cost effects directly.

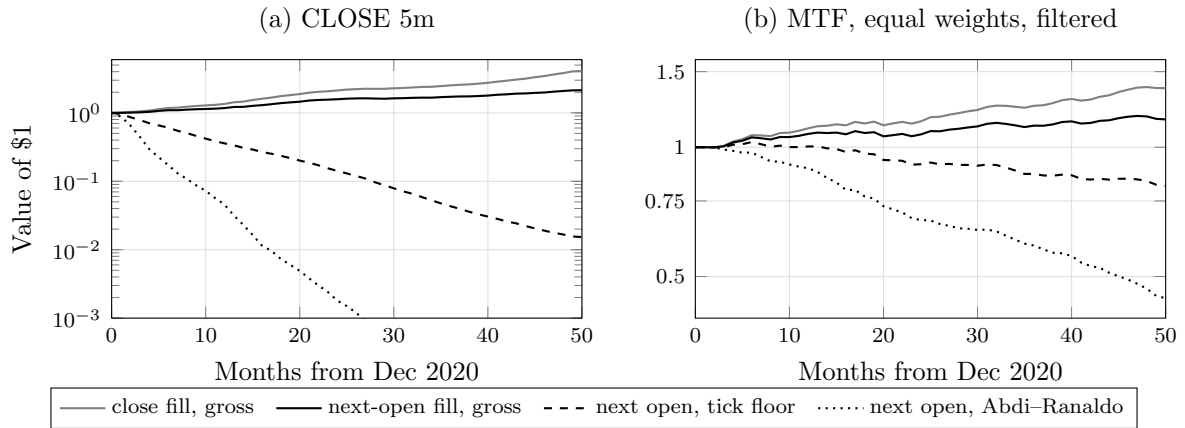


Figure 1: Growth of \$1, log scale, month-end values. Grey to solid: the fill assumption. Solid to dashed: the smallest possible cost. Dashed to dotted: a realistic cost.

Table 4: \$1,000,000 at the start, compounded bar by bar, 2020-12-18 to 2025-01-10. “Vol” is annual volatility; “Max DD” the largest fall from a peak; last column: share of months with a gain.

Strategy	Fill, cost	End value	Total	CAGR	Vol	Sharpe	Max DD	Mo. > 0
CLOSE 5m	close fill, gross	\$4,091,721	+309.2%	+41.6%	3.9%	9.01	−2.2%	98%
	next open, gross	\$2,141,607	+114.2%	+20.7%	3.8%	4.99	−2.6%	92%
	next open, tick floor	\$15,298	−98.5%	−64.4%	3.9%	−27.25	−98.5%	0%
	next open, AR	\$4	−100.0%	−95.3%	3.9%	−79.05	−100.0%	0%
	next open, 0.87 bps	\$3,305	−99.7%	−75.6%	3.9%	−37.25	−99.7%	0%
MTF, equal, sign	close fill, gross	\$2,157,954	+115.8%	+20.9%	3.1%	6.24	−2.0%	96%
	next open, gross	\$1,552,316	+55.2%	+11.5%	3.1%	3.61	−2.3%	84%
	next open, tick floor	\$309,890	−69.0%	−25.1%	3.1%	−9.53	−69.3%	0%
	next open, AR	\$19,949	−98.0%	−62.0%	3.2%	−30.98	−98.0%	0%
	next open, 0.87 bps	\$188,775	−81.1%	−33.8%	3.1%	−13.53	−81.3%	0%
MTF, equal, filtered	close fill, gross	\$1,373,183	+37.3%	+8.2%	2.9%	2.76	−2.9%	72%
	next open, gross	\$1,160,997	+16.1%	+3.8%	2.9%	1.32	−4.1%	62%
	next open, tick floor	\$813,564	−18.6%	−5.0%	2.9%	−1.78	−21.2%	34%
	next open, AR	\$445,066	−55.5%	−18.1%	2.9%	−6.89	−55.6%	4%
	next open, 0.87 bps	\$725,769	−27.4%	−7.6%	2.9%	−2.76	−29.3%	28%
MID 5m	close fill, gross	\$137,618	−86.2%	−38.7%	6.7%	−7.35	−86.3%	10%
	next open, gross	\$336,381	−66.4%	−23.6%	6.6%	−4.14	−66.6%	12%
	next open, tick floor	\$1,527	−99.8%	−79.9%	6.6%	−24.77	−99.8%	0%
	next open, AR	\$0	−100.0%	−97.8%	6.6%	−58.64	−100.0%	0%
	next open, 0.87 bps	\$287	−100.0%	−86.7%	6.6%	−31.15	−100.0%	0%

Table 5: Out of sample, January 2022 – January 2025 (3.01 years): MTF, learned weights, filtered.

Fill, cost	End value	Total	CAGR	Vol	Sharpe	Max DD	Mo. > 0
close fill, gross	\$1,191,641	+19.2%	+6.0%	2.9%	2.07	−3.0%	70%
next open, gross	\$1,069,992	+7.0%	+2.3%	2.9%	0.81	−3.8%	68%
next open, tick floor	\$870,953	−12.9%	−4.5%	2.9%	−1.61	−14.6%	38%
next open, AR	\$616,553	−38.3%	−14.8%	2.9%	−5.55	−38.4%	11%
next open, 0.87 bps	\$814,363	−18.6%	−6.6%	2.9%	−2.39	−19.6%	30%

CLOSE 5m. Before costs the book is excellent: \$4,091,721 with the close fill, \$2,141,607 with the next-open fill, and volatility of only 3.8% because it is long about half the stocks and short the rest, so market moves cancel. The book changes 0.95 units of position per bar. At the median tick floor that alone costs about 0.450 bps per bar, close to its whole gross return, and the book ends at \$15,298. A high Sharpe ratio and a near-total loss are both correct: many tiny bets, each costing more to place than it earns.

MTF, filtered. Trading far less, it keeps more: \$1,160,997 gross, \$813,564 after the tick floor (−19%, worst drawdown −21.2%), \$445,066 after Abdi–Ranaldo costs. Out of sample with learned weights (Table 5) it ends at \$1,069,992 before costs and \$870,953 after the tick floor.

MID 5m uses the midpoint of the range as its price and loses money before any cost (\$336,381 with the next-open fill). MID’s apparent accuracy comes from a known term in the midpoint return, not from information.

5.3 Statistical confidence

Table 6: 95% confidence intervals, moving-block bootstrap over trading days (Künsch, 1989) (20-day blocks, 1,000 draws), next-open fill.

Strategy	Gross Sharpe	Breakeven (bps)	After the tick floor (%)	
			CAGR	Max DD
CLOSE 5m	[4.10, 5.97]	[0.083, 0.122]	[−66.0, −62.8]	[−98.7, −98.2]
MTF, equal, sign	[2.66, 4.66]	[0.134, 0.233]	[−27.5, −22.8]	[−72.8, −65.0]
MTF, equal, filtered	[0.32, 2.39]	[0.068, 0.504]	[−7.7, −2.1]	[−28.7, −10.8]
CLOSE 4h only	[0.19, 2.87]	[0.075, 1.061]	[−2.9, 2.0]	[−12.9, −2.4]
MTF, learned, filtered, 2022+	[−0.40, 1.97]	[−0.108, 0.515]	[−7.7, −1.5]	[−22.2, −7.3]

The loss is not noise. Days are resampled in blocks to keep intraday and short-run dependence. After the tick floor the CAGR interval lies below zero for every book except the 4-hour model alone, which cannot be told apart from breakeven. The filtered book’s breakeven interval is wide (0.037 to 0.516 bps across block lengths of 5, 20 and 60 days), so small differences between combined versions are not significant.

The gross edge is not an artefact of search. Trying many versions inflates the best Sharpe ratio (Harvey et al., 2016). The deflated Sharpe ratio of Bailey and López de Prado (2014) corrects for the number of trials and for non-normal returns. Counting every configuration run on these data, 102 including an earlier specification search and the robustness checks, the best gross Sharpe (6.26) has a deflated Sharpe probability of 0.978. And selection cannot explain a loss: 100% of the 46 configurations lose money at the tick floor.

5.4 Risk and data quality

Risk (Table 7). The books have no market exposure to speak of: net exposure is close to zero on average and daily beta is near zero. Losses after costs therefore come from trading, not from the market.

Stale prints (Table 8). 44 thinly traded, high-priced stocks (e.g. AIZ, AVY, AZO, BKNG, CMG, CRL) have many bars in which the price does not move. Unchanged prices followed by a move look like reversals. Without these stocks the gross edge is smaller, especially with the close fill, and every conclusion holds. Interactive Brokers builds bars from filtered trades, so

Table 7: Risk before costs, next-open fill. Net exposure is the average position across stocks (+1 long, -1 short); beta and correlation are against the equal-weight return of all stocks, daily; DD days is the longest time below a previous peak.

Strategy	Net exp.	99th pct net	Gross exp.	Beta	Corr.	Max DD	DD days
CLOSE 5m	0.009	0.61	1.00	-0.001	-0.004	-2.6%	83
MTF, equal, sign	0.014	0.49	1.00	-0.010	-0.039	-2.3%	60
MTF, equal, size	0.001	0.03	0.04	-0.000	-0.008	-0.2%	79
MTF, equal, filtered	0.030	0.45	0.97	-0.006	-0.024	-4.1%	191
MTF, learned, filtered	0.011	0.45	0.97	-0.004	-0.018	-3.8%	267
CLOSE 4h only	-0.010	0.55	0.39	0.002	0.014	-3.9%	367
MID 5m	0.006	0.94	1.00	-0.014	-0.024	-66.6%	1018

Table 8: Without the 44 stocks with more than 5% of bars where $H = L$ or more than 1% with zero volume. Breakeven in bps; total return after the tick floor, next-open fill.

Strategy	B/E close fill		B/E next open		Total, tick floor	
	All	Excl.	All	Excl.	All	Excl.
CLOSE 5m	0.190	0.145	0.103	0.088	-98%	-99%
MTF, equal, sign	0.318	0.240	0.182	0.162	-69%	-74%
MTF, equal, filtered	0.590	0.377	0.280	0.222	-19%	-23%
MTF, learned, filtered	0.649	0.368	0.309	0.207	-13%	-18%
CLOSE 4h only	0.566	0.522	0.558	0.515	-2%	-4%
MID 5m	-0.243	-0.206	-0.133	-0.123	-100%	-100%

C_t may not be the last print on the consolidated tape; tick data would sharpen the fill-timing result.

Exploratory, not a strategy. Looking by time of day after the results were known, the filtered book’s breakeven is highest late in the afternoon (1.59 bps in 15:00–15:30). The comparison was not a planned test and should be checked on new data before it is believed.

6 What would make it tradable

The gross edge is real, but it is the spread. A trader who crosses the spread pays it; a trader who posts limit orders at the bid after a down bar and at the ask after an up bar would earn it. Providing liquidity is a different business. A resting order fills only when the market trades through it, and the orders that fill are disproportionately the ones the market was about to move against (adverse selection); fills are uncertain, queue position matters, and exchange rebates and fees change the numbers. None of this can be measured without quote and order-book data, so we do not claim that a passive version would be profitable. What the analysis does show is the size of the target: the book can afford to pay at most 0.280 bps per unit of position change, including adverse selection, to break even, while a market order pays at least 0.47 bps at the median stock.

A follow-up test on the same bars. After this analysis we ran the simplest passive version, under a specification fixed before the run and kept in the repository (`docs/spec`). Each order rests at the signal bar’s close and fills, at that price, only if the next bar trades strictly through it; unfilled orders are cancelled. For the filtered book in 2022–2025, 83% of orders fill, the price moves -4.6 bps against an order over the bar in which it fills and +25.3 bps in its favour over the bar in which it does not, and the net Sharpe ratio falls to -15.1, against -2.3 with market orders at the tick floor. Breaking even would need a rebate of 3.47 bps per unit traded. On

these bars, resting at the last price does not collect the spread; whether better queue position or faster cancellation would is a question for order-book data.

7 Limitations

The verdict rests on bar data and on spread charges alone; each limit below states which way it moves the result.

Costs from bars, not quotes. The tick floor is exact but is a lower bound; Abdi–Ranaldo and Corwin–Schultz are estimates with known biases. Quote data would replace them.

No market impact. Only the spread is charged. An equal-weight book in 431 stocks trading every five minutes would move prices at any meaningful size, so the true cost is higher. Omitting impact therefore makes the negative verdict conservative.

Selection. The sample contains stocks with data through 2025. The set of versions was chosen after earlier results on the same data; the deflated Sharpe ratio corrects the gross result for this but a fresh set of stocks would be cleaner.

8 Conclusion

A 5-minute reversal signal with a Sharpe ratio of 9.01 in a standard backtest loses money for any trader who uses market orders. About half of its edge is the assumption of a fill at the signal bar’s own close: Proposition 1 shows that the gap between the two fills equals position change times the jump to the next print, and the measured jump is bid–ask bounce. The rest is smaller than the smallest possible cost of a trade, and resting limit orders at the last close, tested in a follow-up fixed in advance, do worse because their fills are adversely selected. The edge grows with the spread, combining timeframes and filtering improve the edge per trade 1.8 to 2.7 times but not enough, and trading only cheap stocks does not help once costs are measured without estimation error. The practical lessons are to fill at the first print after the signal, to charge costs per stock with an exact lower bound, and to judge a high-turnover strategy by its breakeven cost, not its Sharpe ratio.

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